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Data Analytics with Tableau

Assignment

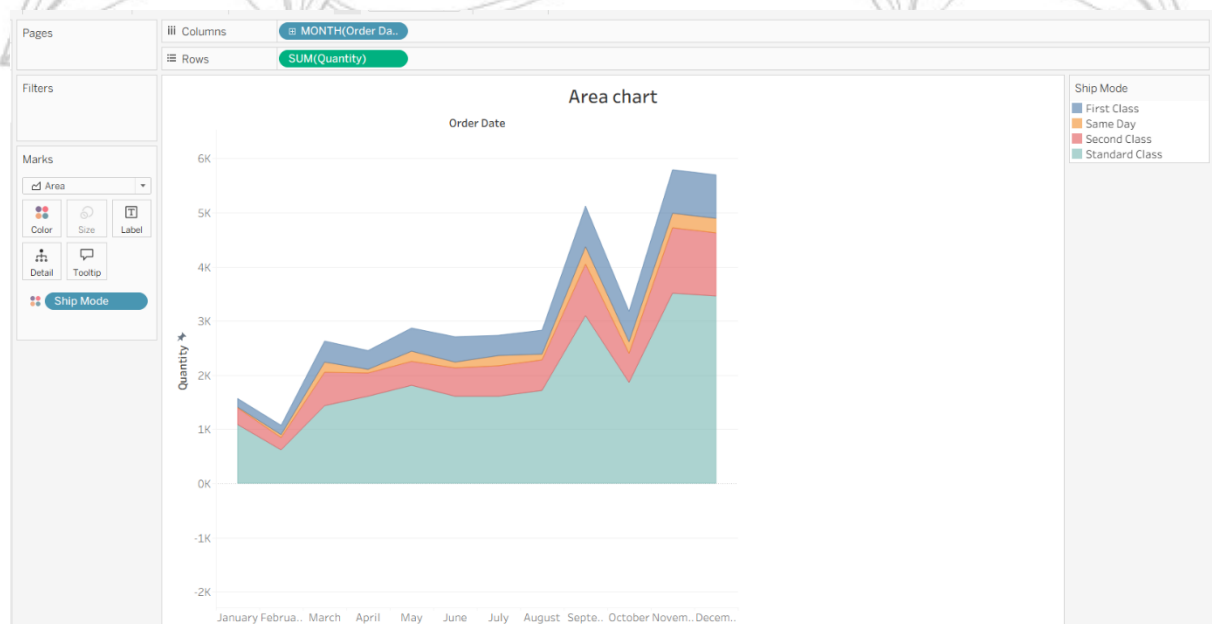
QUESTION: Prepare the following charts

Sheet - 01 – Area chart

Steps:

1. Drag **Order Date** to the column shelf and right-click it to select **month**.
2. Drag **Quantity** to the Rows shelf.
3. Drag **Ship Mode** to the **Color** icon on the marks card.
4. On the marks card, change the drop-down menu selection from “Automatic” to **Area**.

Output

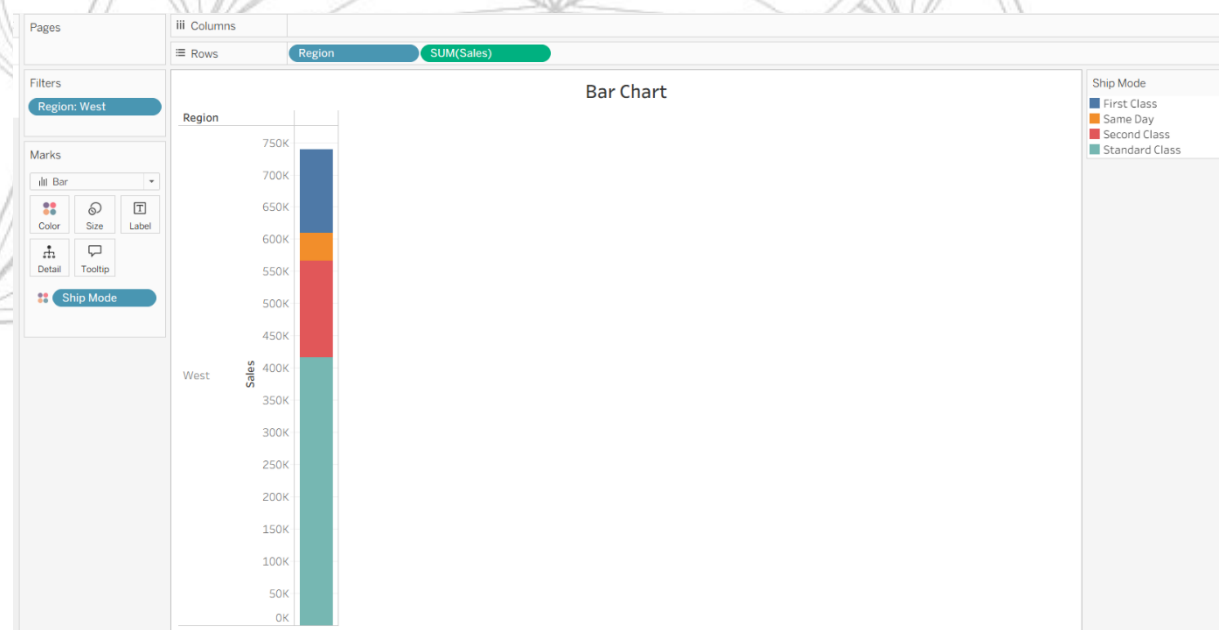


Sheet – 02 – Bar Chart

Steps:

1. Drag **Sales** to the **Rows Shelf**.
2. Change **Marks** to **Bar** using the marks card drop-down menu.
3. Drag **Ship Mode** to **Color** on the marks card to add color coding.
4. Drag **Region** to the **Rows Shelf** (place it to the left of sales).
5. Drag **Region** to the **Filters Shelf** and select only **West**.

Output:



Sheet – 03 – Box plot

Steps:

1. Drag **Segment** to the **Columns Shelf** and **Discount** to the **Rows Shelf**.
2. Change **Aggregation** by clicking Discount in the Rows shelf and selecting **Measure (Average)**.
3. Drag **Region** to the **Columns Shelf** and drop it to the right of Segment.
4. Click “**Show Me**” in the toolbar and select the **Box – and – Whisker Plot**.
5. Drag **Region** from the Marks card back to the **Columns shelf** (right of segment).
6. **Disaggregate Data** by going to the Analysis Menu and unchecking **Aggregate Measures**.
7. Click “**Swap Rows and Columns**” in the toolbar to switch to a horizontal view.

Output:

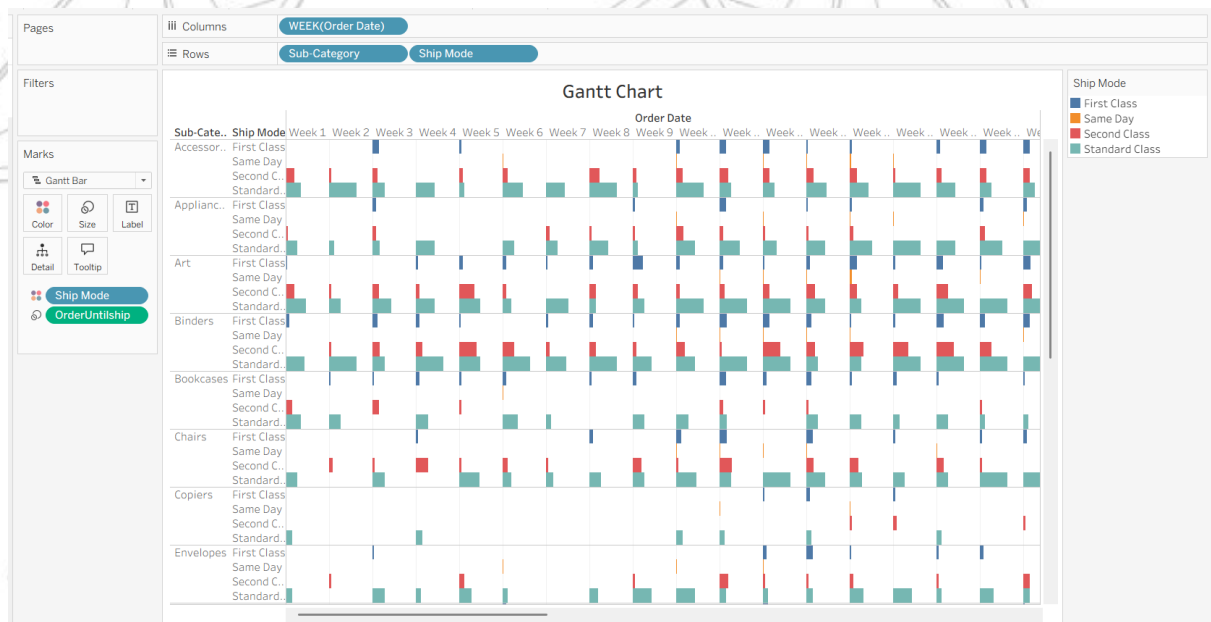


Sheet – 04 – Gantt chart

Steps:

1. Drag **Order Date** to the **Columns Shelf**.
2. Click the drop-down on the Order Date pill and select More to **Week Number** to display a weekly timeline.
3. Drag **Sub-Category** to the **Rows Shelf**, then drag **Ship Mode** to the right of it.
4. In the Marks card drop-down menu, select **Gantt Bar**.
5. Go to **analysis** to **Create Calculated Field**, name it “OrderUntilShip,” and use the formula: DATEDIFF (‘day’, [Order Date], [Ship Date]).
6. Drag the new the **OrderUntilShip** field onto **Size** in the Marks card.
7. Right-click the OrderUntilShip pill on the marks card and select **Measure to Average**.
8. Drag **Ship Mode** to the **color** Shelf on the Marks card to distinguish the bars.

Output:

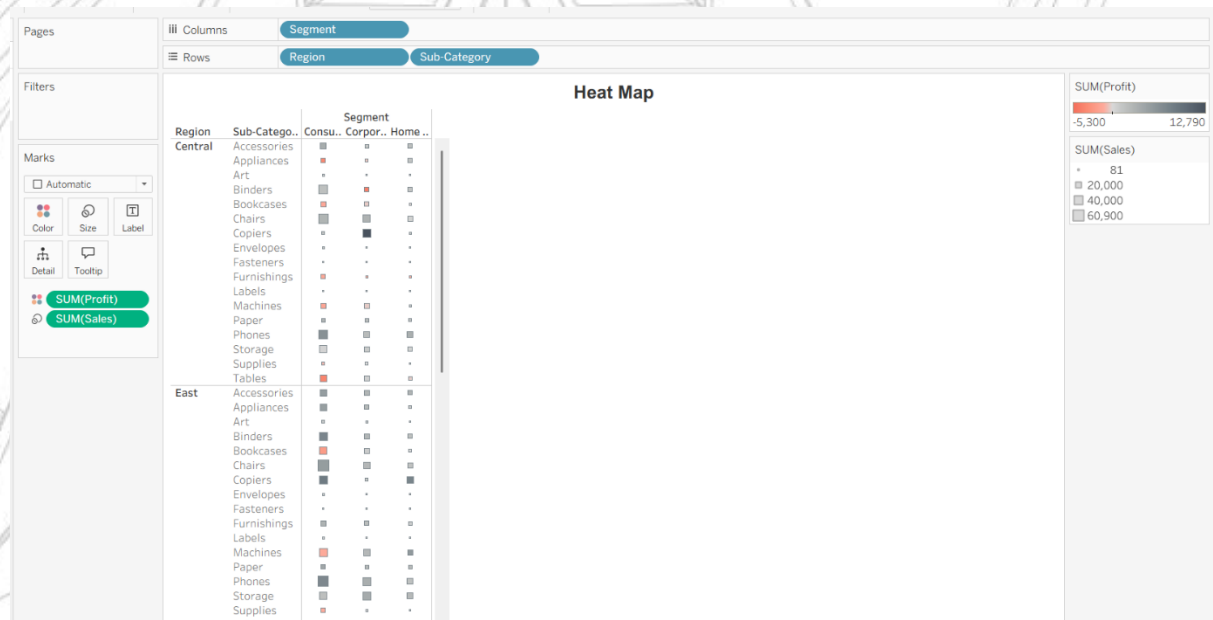


Sheet – 05 – Heat Map

Steps:

1. Drag **Segment** to columns and both **Region** and **Sub-Category** to Rows.
2. Drag the **Profit** measure onto the **Color** mark and set the Palette to **Red-Black Diverging**.
3. Drag the **Sales** measure onto the **Size** mark and adjust the slider to ensure boxes fit the cells.
4. Click **Color** on the marks card and select a **Medium Gray** border for visual clarify.

Output:



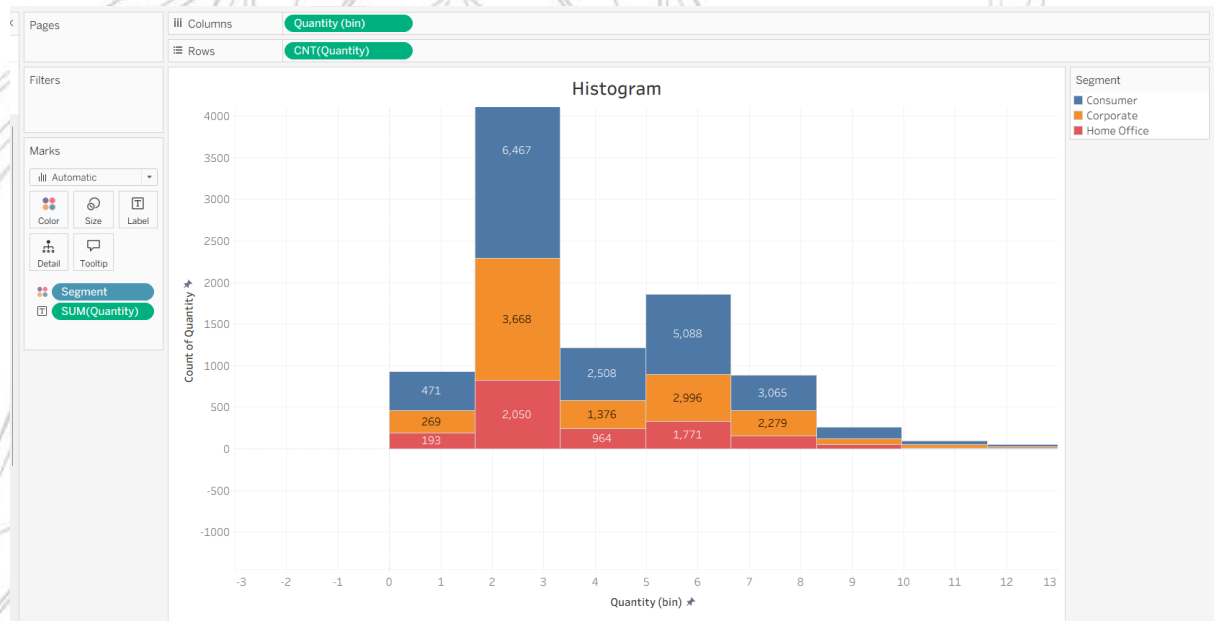
Sheet – 06 – Histogram

Steps:

1. Drag **Quantity** to the columns Shelf and select the **Histogram** icon from the **Show Me** pane to automatically create bins and the frequency count.
2. Drag the **segment** dimension onto the **Color** mark to create the stacked bar effect (Consumer, Corporate, and Home Office).

3. Hold **Ctrl** and drag the **CNT(Quantity)** pill from the Rows Shelf onto the Label mark.
4. Right-click the **CNT(Quantity)** pill on the Marks card, select **Quick Table Calculation**, and choose **Percent of Total**.

Output:



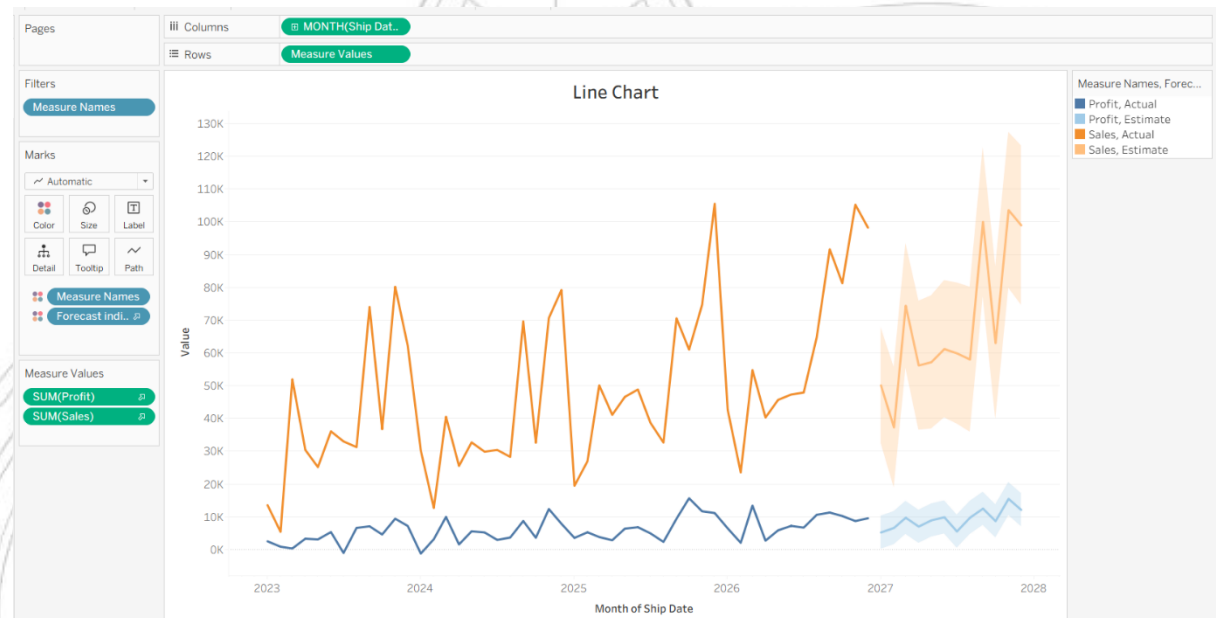
Sheet – 07 – Line Chart

Steps:

1. Drag **Ship Date** to the columns Shelf and change the level to **Month** (the continuous option from the lower section of the dropdown).
2. Drag **Sales** to the Rows Shelf, then drag **Profit** directly onto the existing **Sales axis** in the chart area to create a blended axis.
3. Ensure **Measure Names** is on the **Color** mark to distinguish the Profit and Sales lines.

4. Open the **Analytics pane**, drag **Forecast** into the view, and drop it onto the chart area to predict future trends.

Output:

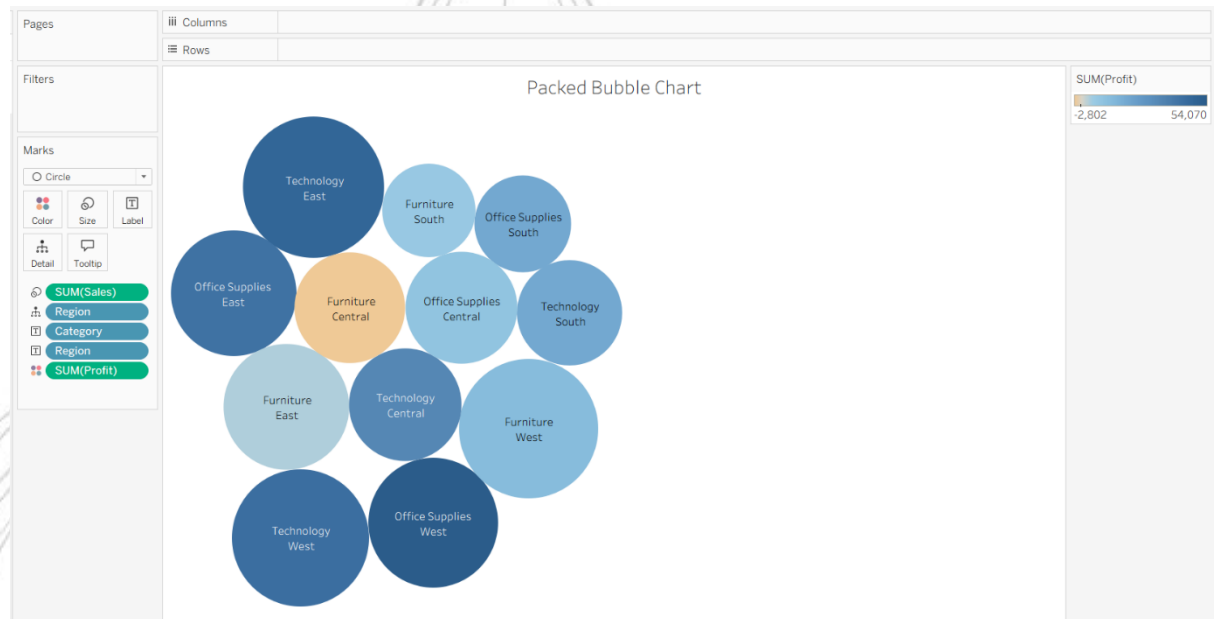


Sheet – 08 – Packed Bubble Chart

Steps:

1. Drag **Category** and **region** to the Columns and Rows shelves, then select the **Packed Bubble** icon from the **Show Me** pane.
2. Drag the **Sales** measure onto the **Size** mark to determine the diameter of each bubble.
3. Drag the **Profit** measure onto the **Color** mark to visualize profitability (e.g., Dark Blue for high profit, orange/Red for loss).
4. Drag both **Category** and **Region** onto the **Label** mark so each bubble identifies its specific data segment.

Output:



Sheet – 09 – Pie Chart

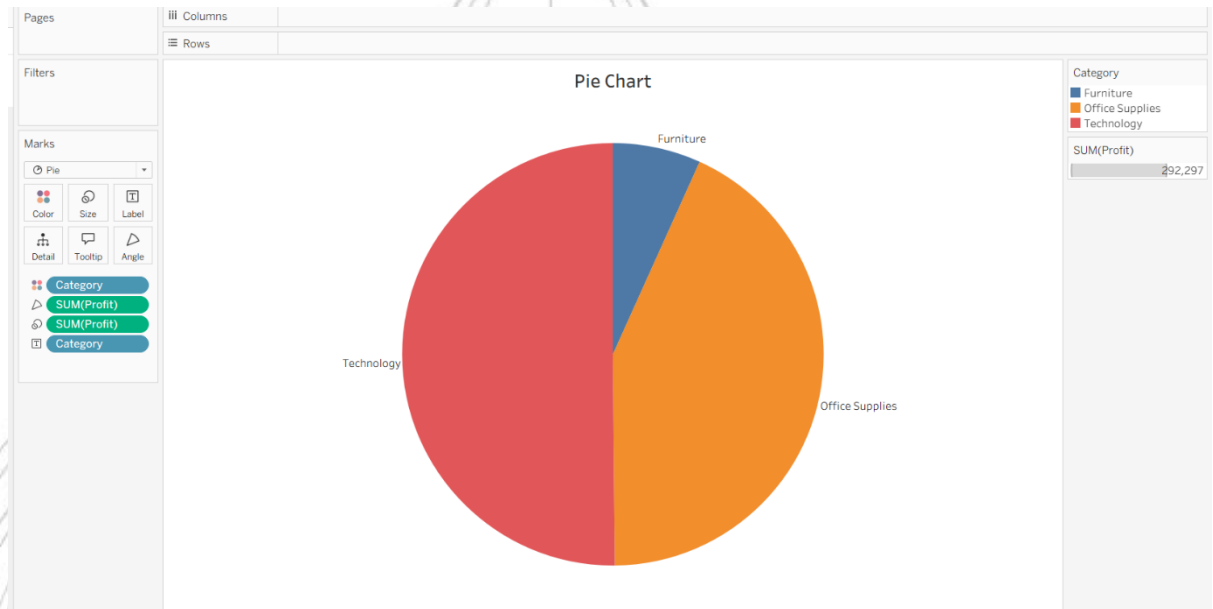
Steps:

1. Drag **Profit** to columns and **Category** to Rows, then select the **Pie chart** icon from **show Me** pane.
2. Ensure the **Category** dimension is on the **Color** mark to create distinct slices for Furniture, Office supplies, and Technology.
3. Drag the **Profit** measure onto the **Angle** mark to determine the proportional size of each slice.
4. Drag the **Category** dimension onto the **Label** mark to display the name of each category next to its slice.

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Output:



Sheet – 10 – Scatter Plot

Steps:

1. Drag the **Sales** measure to the Columns shelf and the **Profit** measure to the Rows shelf.
2. Drag the **Category** dimension to the **Color** mark and the **Region** dimension to the **Detail** mark to break the single point into multiple marks.
3. Open the **Analytics Pane**, drag **Trend Line** onto the view, and select the **Linear** model.
4. Click on the **Shape** mark to select open circles and adjust the **color** transparency to match the output style.

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Output:



Sheet – 11 – Text Table with Percentages

Steps:

1. Drag **Order date** to the Columns shelf, and drag both **Region** and Sub-category to the Rows shelf.
2. Drag the **Sales** measure onto the **Text (ABC)** mark to fill the grid with values.
3. Right-click the **SUM(Sales)** pill in the Marks card and select **Quick Table Calculation > Percent of Total**.
4. Right-click the pill again, select **Compute Using**, and choose **Pane (Down)** to ensure percentages are calculated within each region.

Output:

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Pages

Columns

YEAR(Order Date)

Rows

Region

Sub-Category

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

SUM(Sales)

Text Table With Percentage

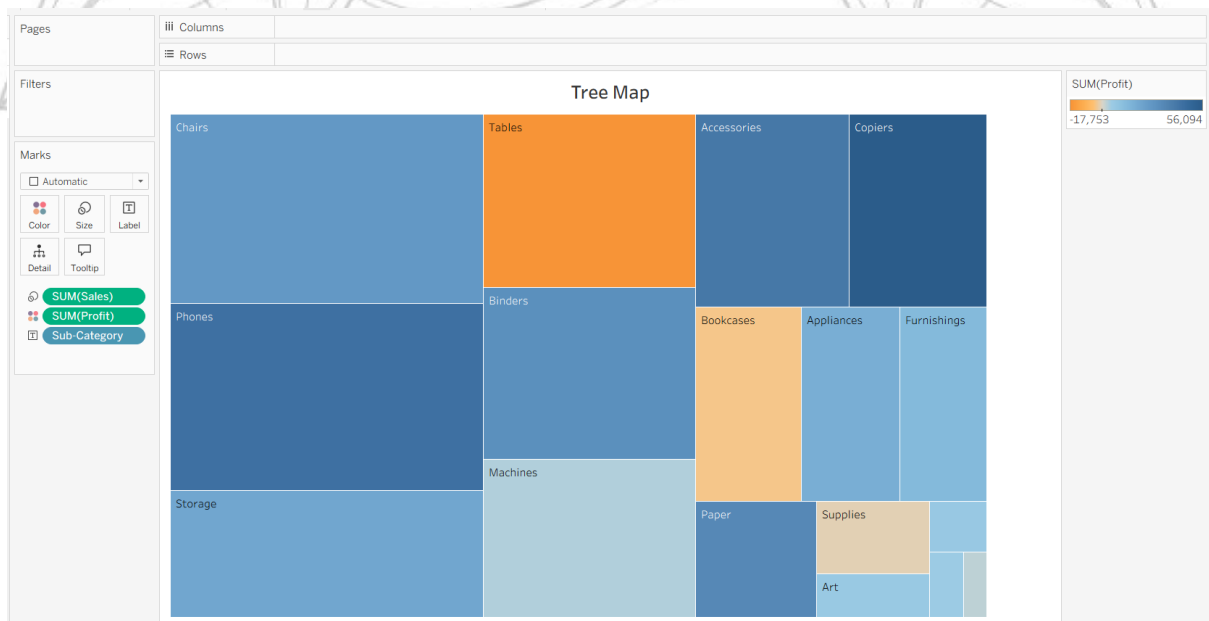
Region	Sub-Catego...	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%
East	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%
	Bookcases	8.02%	12.51%	3.30%	3.42%
	Chairs	16.89%	12.75%	12.70%	14.50%
	Copiers	3.91%	3.16%	10.81%	11.27%
	Envelopes	0.99%	0.56%	0.71%	0.39%
	Fasteners	0.16%	0.14%	0.21%	0.13%
	Furnishings	3.32%	3.46%	5.43%	5.07%
	Labels	0.38%	0.33%	0.47%	0.42%
	Machines	10.81%	9.18%	13.80%	5.46%
	Paper	3.25%	2.53%	2.71%	3.62%
	Phones	16.04%	14.50%	11.76%	16.79%
	Storage	12.07%	9.48%	12.11%	9.48%

Sheet – 12 – Treemap

Steps:

1. Drag **Sub-Category** to columns and sales to rows, then select the **Treemap** icon from the **Show Me** pane.
2. Ensure **Sub-Category** is on the **Label** mark to define the individual rectangles.
3. Drag the **Profit** measure onto the **Color** mark and the **Sales** measure onto the **Size** mark to define rectangle dimensions.

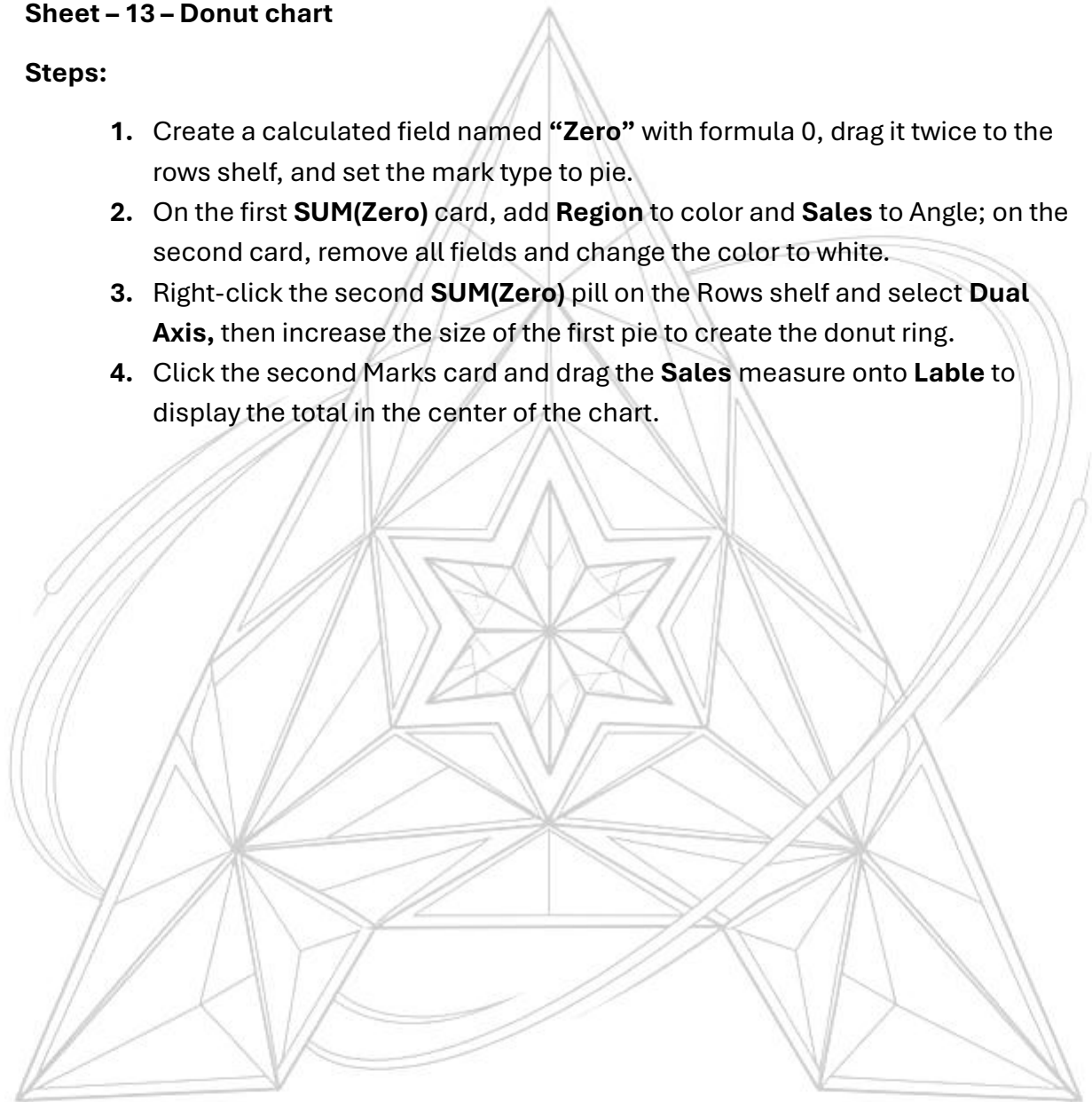
Output:



Sheet – 13 – Donut chart

Steps:

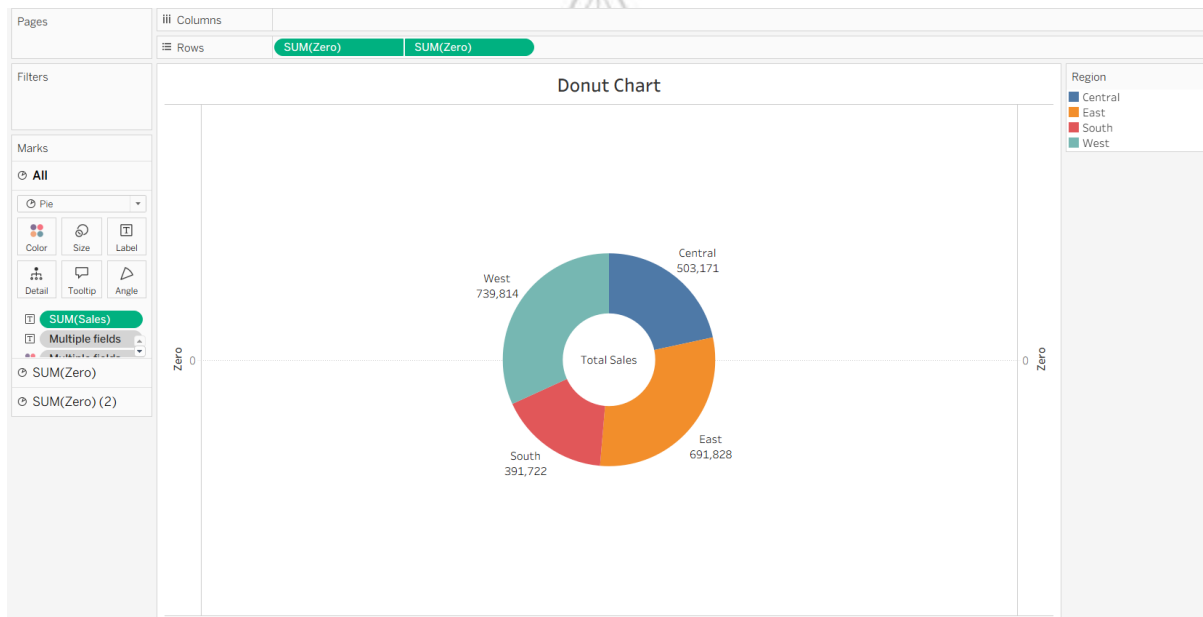
1. Create a calculated field named **“Zero”** with formula 0, drag it twice to the rows shelf, and set the mark type to pie.
2. On the first **SUM(Zero)** card, add **Region** to color and **Sales** to Angle; on the second card, remove all fields and change the color to white.
3. Right-click the second **SUM(Zero)** pill on the Rows shelf and select **Dual Axis**, then increase the size of the first pie to create the donut ring.
4. Click the second Marks card and drag the **Sales** measure onto **Label** to display the total in the center of the chart.



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Output:



Sheet – 14 – Map Chart

Steps:

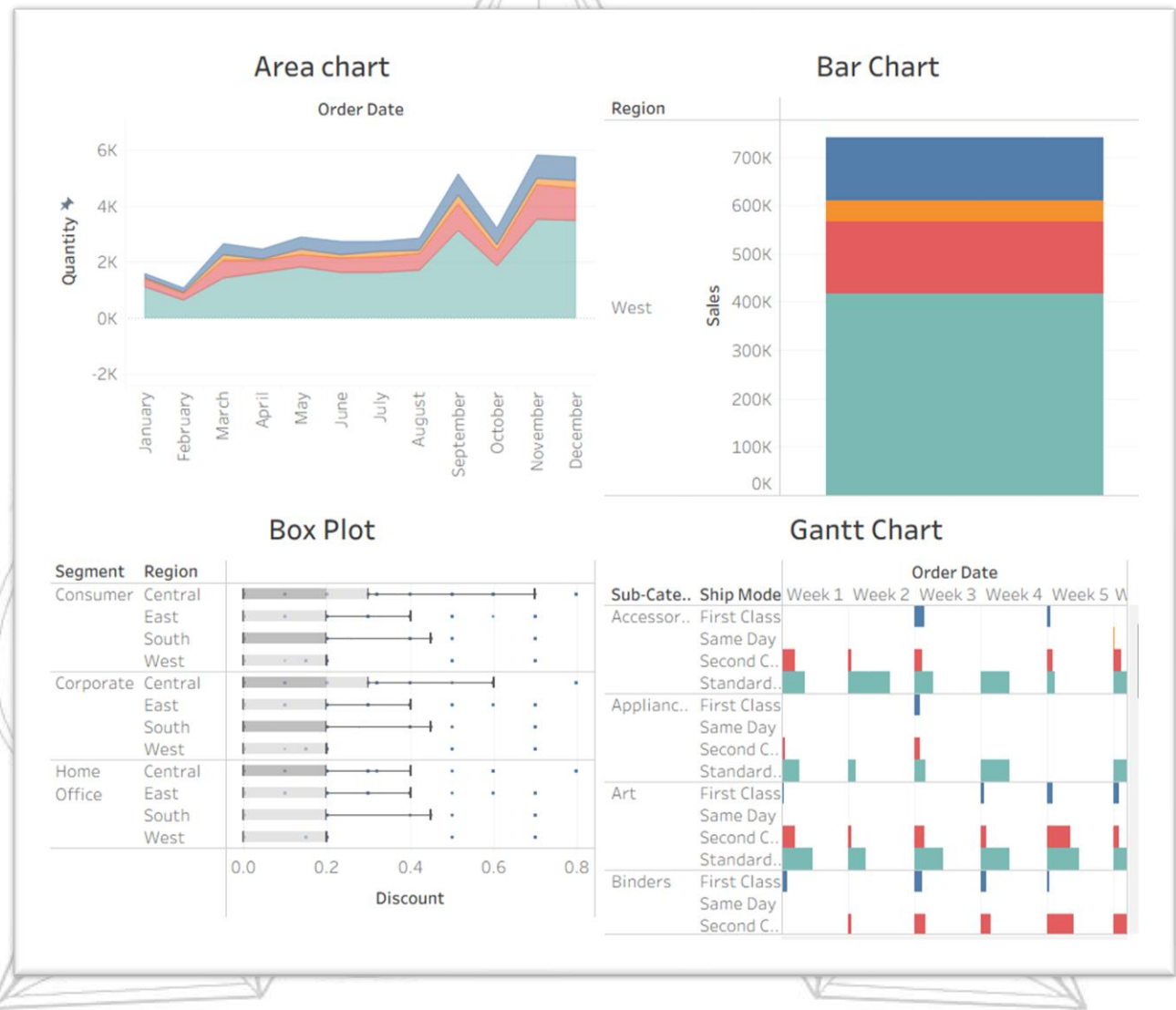
1. Double-click **state** (or drag to the workspace) to automatically generate a map using Latitude and Longitude.
2. Change the mark type to **Map** (Filled Map) and drag the **Sales** measure onto the **Color** mark using a **purple** palette.
3. Drag **State** and **Sales** onto the **Label** mark and add **Region** to **Filters** shelf for interactive analysis.
4. Go to **Map > map Layers** and set the background style to **Outdoors** with a **70%** washout for better readability.

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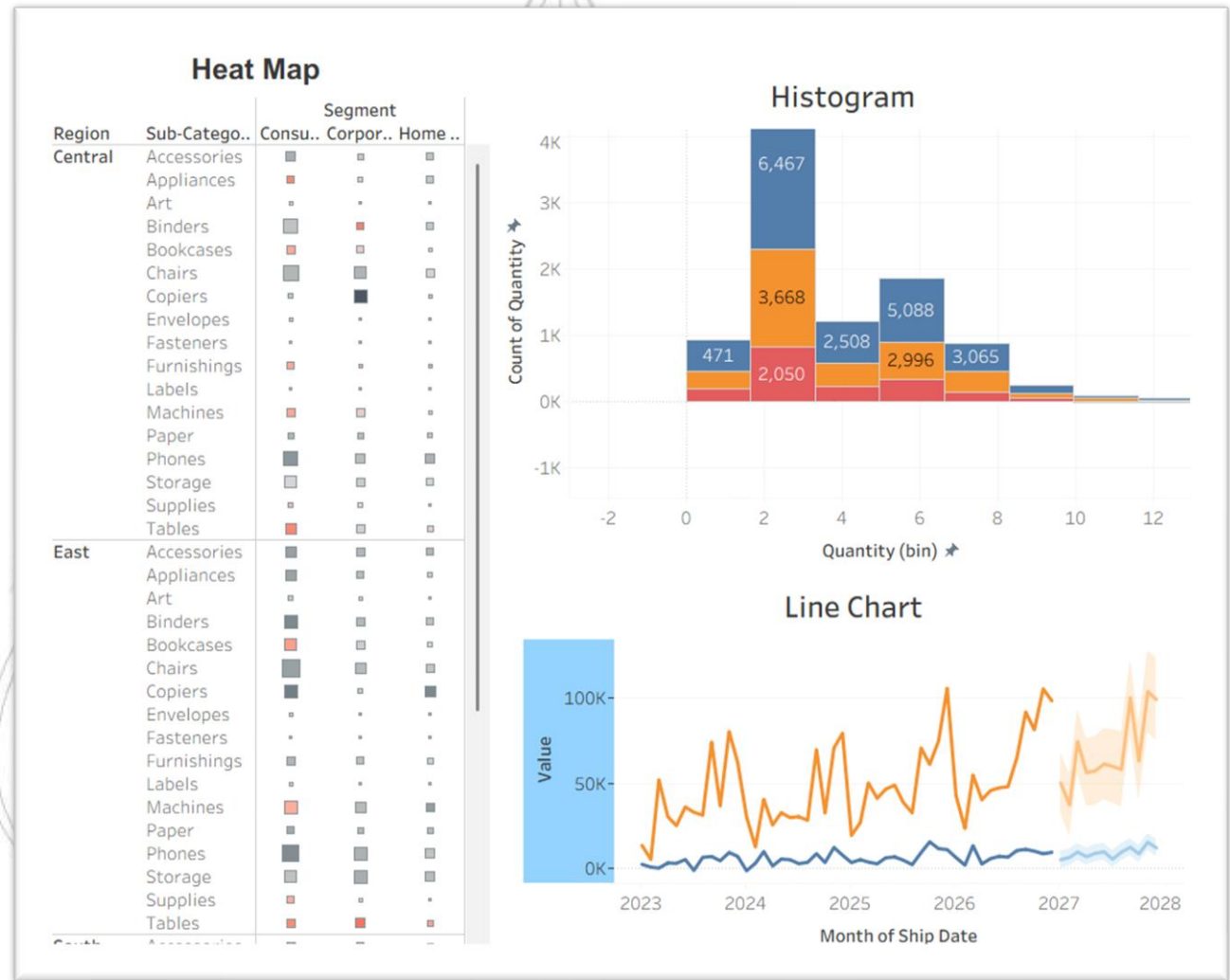
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DASHBOARDS:

Dashboard – 01

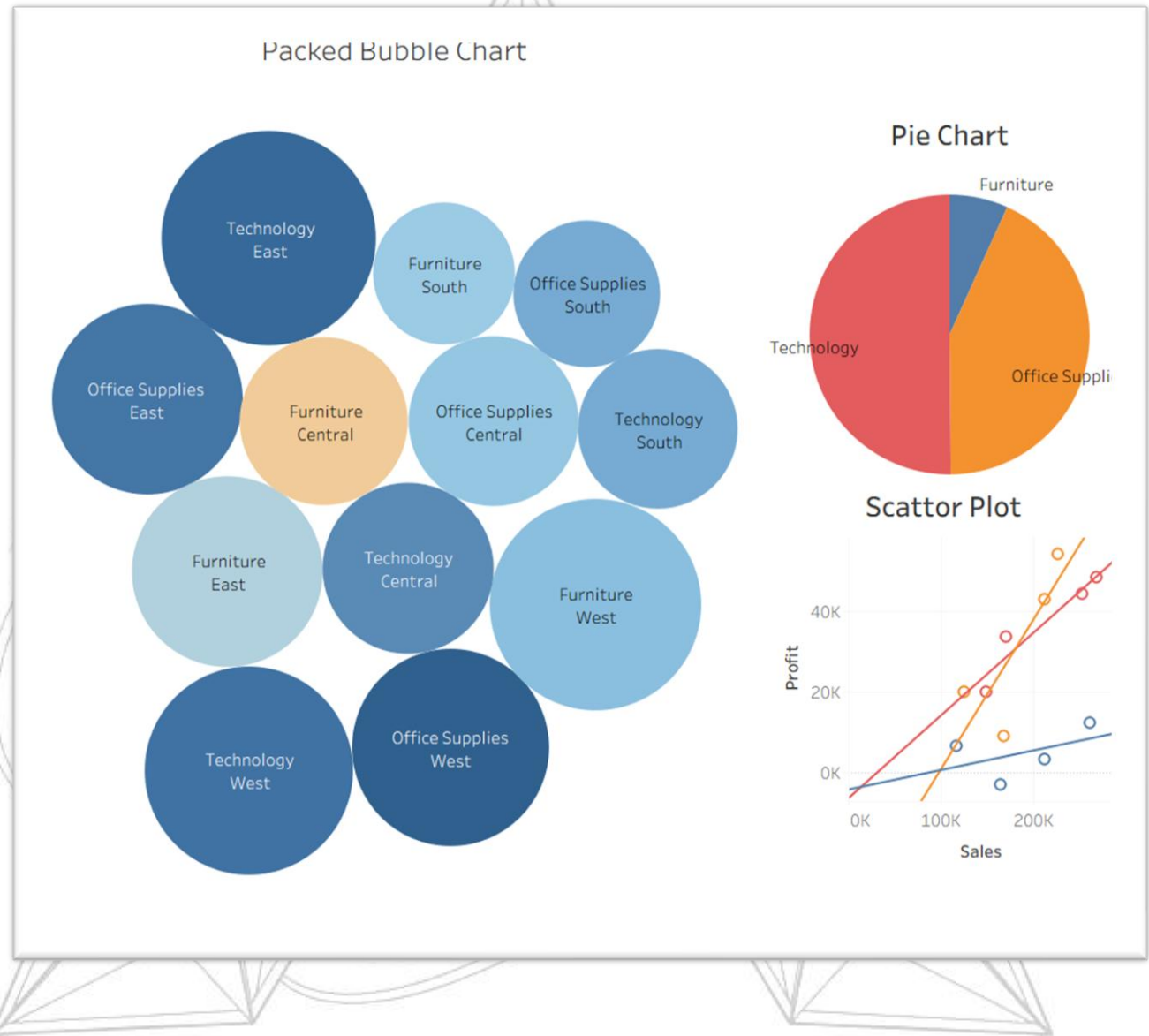


Dashboard – 02



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Dashboard – 03



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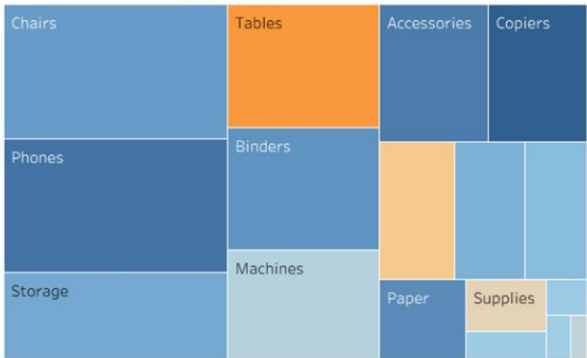


Dashboard – 04

Text Table With Percentage

Region	Sub-Catego...	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
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	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%

Tree Map

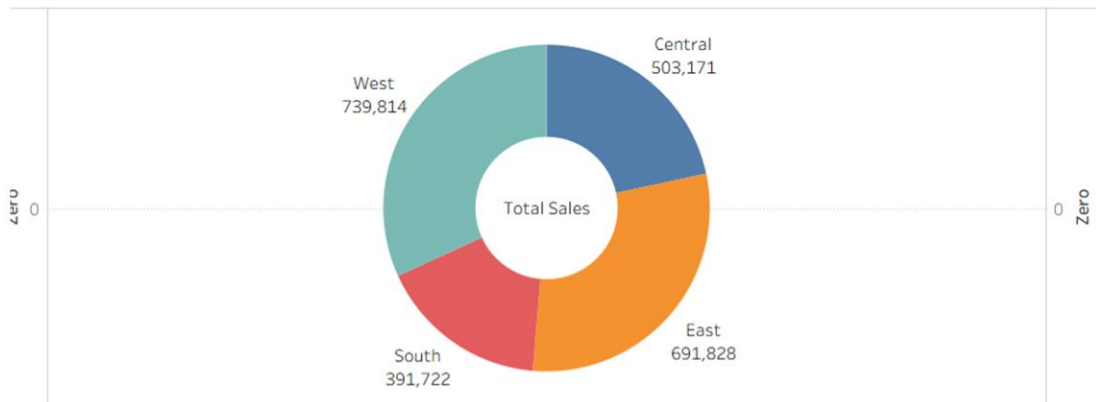


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Dashboard – 05

Donut Chart



Map Chart



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